Sultan Qaboos University

College of Engineering

Petroleum and Chemical Engineering Department

Chemical Reaction Engineering

CHPE4512

Production of Cumene

Memo 5

Section 02

Group 03

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Chapter 5: Energy Balance and Reactor Stability Analysis

5.1.1 Energy Balance Implementation

To represent the thermal behavior of the Cumene packed-bed reactor more realistically, the reactor model developed in Memo 4 was extended by adding an energy balance to the material balances. In Memo 4, the reactor was treated as isothermal, which means that the temperature was assumed to remain constant throughout the catalyst bed. Although this assumption simplifies the analysis, it does not reflect the actual behavior of the reactor because the reactions are exothermic and heat is exchanged continuously with the cooling medium. Therefore, a non-isothermal model was developed in the present work to predict the temperature variation along the reactor bed.

The reactor temperature was described using a differential energy balance written with respect to catalyst weight, W. The energy balance includes the heat released by the chemical reactions, the heat removed through the reactor wall to the coolant, and the heat capacity flow of the reacting mixture. For the Cumene reactor, the desired reaction is the formation of Cumene, while the secondary reaction leads to the formation of DIPB. Since both reactions are exothermic, they release heat and contribute to an increase in reactor temperature. At the same time, heat is removed by the external coolant, which helps control the temperature rise inside the reactor.

The energy balance used in this work is written as:

$$\frac{dT}{dW} = \frac{-r_1 \Delta H_1 - r_2 \Delta H_2 + U.a.(T_a - T)/\rho_{bulk}}{\sum_i F_i C_{p,i}}$$

where r_1 and r_2 are the rates of the main and side reactions, respectively, ΔH_1 and ΔH_2 are the heats of reaction, U is the overall heat transfer coefficient, a is the heat transfer area per reactor volume, T_a is the coolant temperature, ρ_{bulk} is the catalyst bulk density, F_i is the molar flow rate of species i, and $C_{p,i}$ is the heat capacity of species i.

For the present simulation, the inlet temperature was 628.1 K, the coolant temperature was 618 K, and the overall heat transfer coefficient was taken as 200 W/ (m^2.K). In addition, the heat capacities of propylene, benzene, Cumene, DIPB, and the inert component were included in the model as constant values. These parameters were incorporated directly into the energy balance and used to predict the temperature profile along the catalyst bed.

Assumptions

- The reactor operates at steady-state conditions.
- The reactor behaves as a plug flow packed-bed reactor.
- The reactor is non-isothermal, and the temperature varies with catalyst weight.
- The heat capacities of all species are assumed to remain constant over the operating range.
- The overall heat transfer coefficient, U, is assumed to be constant.
- The coolant temperature is assumed to be constant along the reactor length.
- Axial heat dispersion is neglected.
- Radial temperature gradients are neglected.
- Only the main reaction to Cumene and the side reaction to DIPB are included in the thermal model.
- The physical properties used in the model are assumed to be suitable for the selected operating conditions.

The heat capacities used in the simulation (assumed constant and based on feed temperature of 355 ℃) are as follows:

Table 1: Heat Capacities [1]

Species	Cp (J/mol·K)
Propylene, A	111.11
Benzene, B	164.76
Cumene, C	343.90
DIPB, D	412.50
Inert, I	129.47

Including the energy balance is essential because temperature directly affects reaction rate, conversion, selectivity, and reactor safety. In the Cumene reactor, neglecting the temperature variation would hide important thermal behavior, especially hotspot formation. Therefore, the non-isothermal model provides a more reliable basis for reactor analysis than the isothermal model used previously.

5.1.2 Temperature Profile Analysis

After incorporating the energy balance into the reactor model, the reactor temperature profile was evaluated as a function of catalyst weight under the selected base-case conditions. The resulting profile is shown in Figure 1. The figure indicates that the reactor temperature increases sharply from the inlet value of 628.1 K and reaches a maximum value of approximately 875.8 K at a catalyst weight of about 1.97 kg per tube. This point represents the hotspot of the reactor. After reaching this maximum value, the temperature decreases gradually along the remaining section of the catalyst bed.

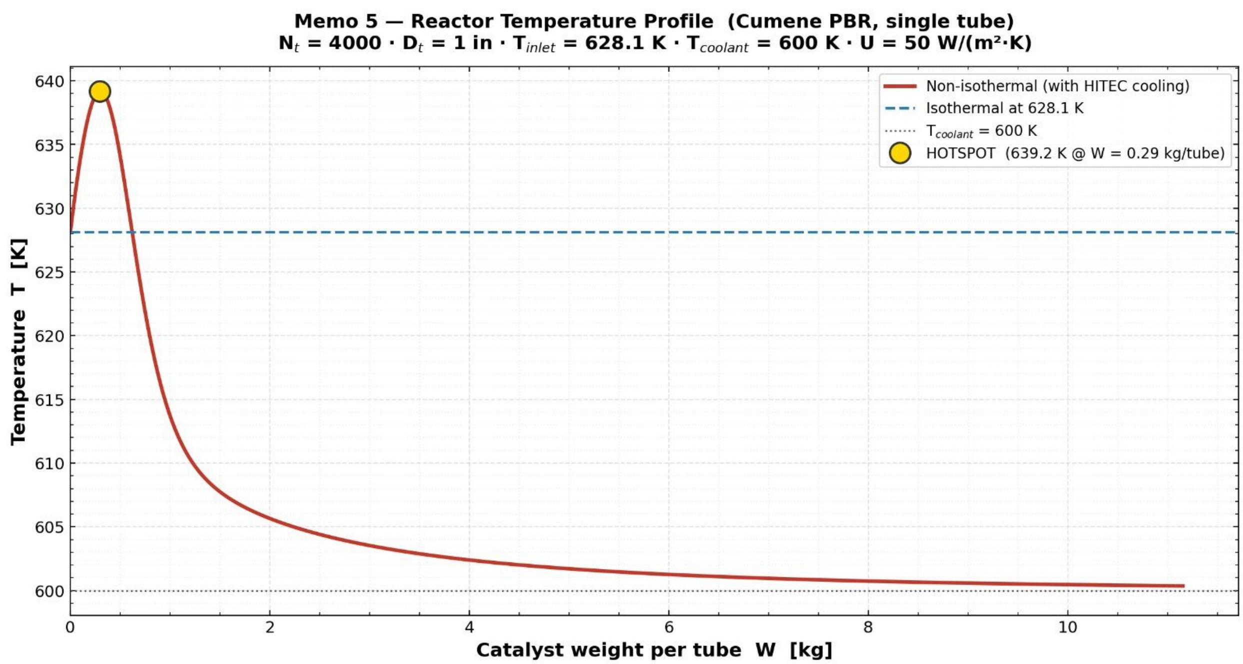

Figure 1: Temperature profile along catalyst weight

The sharp rise in temperature near the reactor entrance indicates that heat generation from the exothermic reactions is dominant in the front section of the bed. At low catalyst weight, the reactant concentrations are still high, and therefore the reaction rate is also high. Under these conditions, the heat released by reaction is greater than the heat removed by the coolant, which causes a rapid increase in temperature. As the reactants are consumed, the reaction rate decreases and the rate of heat generation becomes lower. Since cooling continues along the reactor, the temperature begins to decrease after the hotspot.

Figure 1 also compares the non-isothermal and isothermal cases. In the isothermal model, the reactor temperature remains constant at 600 K throughout the bed. In contrast, the non-isothermal model predicts a large temperature increase followed by a gradual decrease. This comparison clearly shows that the isothermal assumption is not sufficient to describe the real thermal behavior of the Cumene reactor under the selected conditions. Most importantly, the isothermal model does not capture the hotspot, which is one of the most critical features in the analysis of exothermic packed-bed reactors.

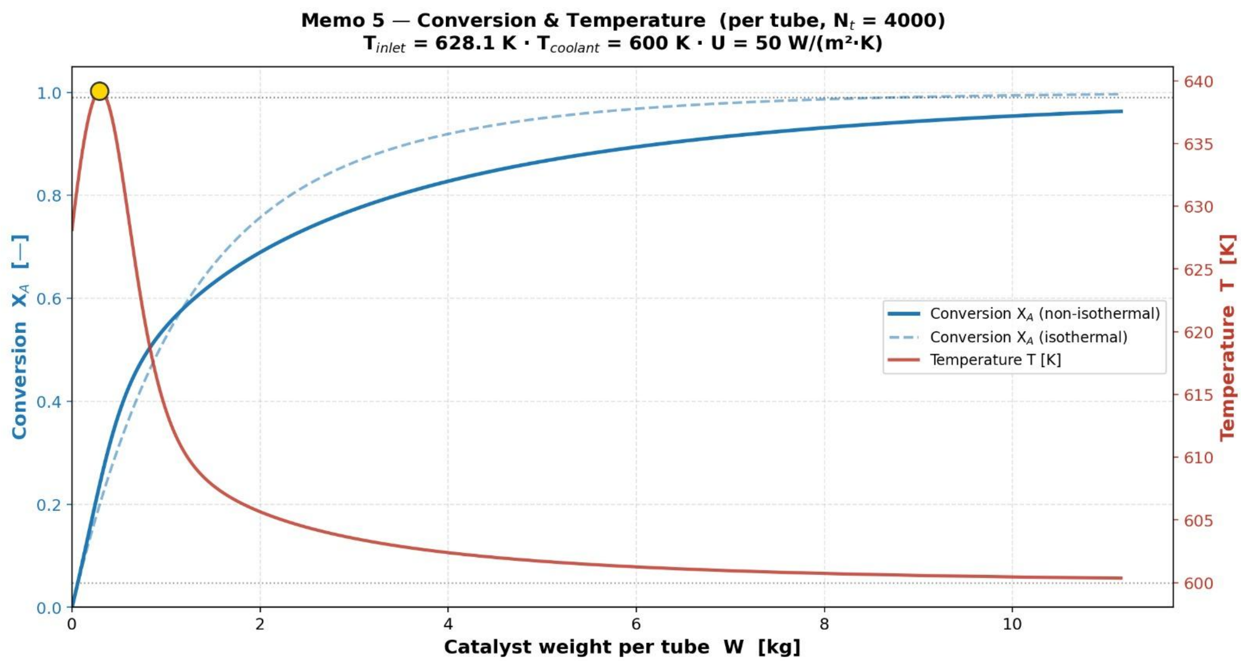

Figure 2: Conversion Versus Catalyst weight per tube

Figure 2 shows that the reactor temperature increases slightly from 628.1 K at the inlet to a small hotspot of about 640 K near the entrance of the bed. This happens because the reaction is fastest at the inlet, where the reactant concentrations are highest, so heat generation is also highest in this region. After the hotspot, the temperature decreases gradually and approaches the coolant temperature of 600 K. This means that, after the first part of the reactor, heat removal by the coolant becomes stronger than the heat generated by reaction. As the reactants are consumed, the reaction rate becomes lower, so less heat is produced. The graph also shows that conversion increases with catalyst weight in both the non-isothermal and isothermal cases. However, the non-isothermal conversion is lower than the isothermal conversion over most of the bed. Although there is a small temperature rise near the inlet, the reactor temperature later drops below the isothermal value, which reduces the reaction rate and slows down conversion.

5.1.3 Selectivity vs. Operating Conditions

The effect of changes in the temperature of the cooling medium (T_{cool}) and the inlet temperature (T_0) on the selectivity towards Cumene in the non-isothermal mode is evaluated in this part. Unlike Memo 4, the non-isothermal mode of operation is defined by the heat balance (dT/dW) computed in Part 1.1. The selectivity towards Cumene in the reactor is basically dependent on the resulting temperature profile because the activation energy for the side reaction (E_{a2} = 120.0 kJ/mol) is higher compared to the activation energy for the main reaction (E_{a1} = 104.2 kJ/mol). With a constant outlet conversion $X_A = 0.99$, two parameter sweeps are performed: in Case A, T_0 changes from 610 to 638 K at a constant T_{cool} = 600 K; in Case B, T_{cool} changes from 580 to 600 K at a constant T_0 = 628 K (design inlet).

Thermophysical Data

Table 2: Heat Capacity and Reaction Enthalpy Data

Species	Cp [J/(mol·K)]	ΔH^{Rx} [kJ/mol]	Reference	Notes
Benzene (A)	164.76	-	Memo 2, Group 03	From handwritten data
Propylene (B)	111.11	-	Memo 2, Group 03	From handwritten data
Cumene (C)	343.90	-1 13.0	Smith & Van Ness	Rxn 1: A+B→C
DIPB (D)	412.50	-1 16.0	Smith & Van Ness	Rxn 2: C+B→D
Propane (I)	45.0	-	NIST WebBook	Inert

Geometry of the reactor tube (N_t = 4000, D_{tube} = 0.05 m, L = 10.75 m) was employed to calculate the heat transfer coefficient: aw = π × 0.05 × 10.75 × 4000/44,610 = 0.151 m²/kg, yielding U_a = 100 × 0.151 ≈ 15 W/(kg·K). In order to take into account possible fouling effects, a conservative value of U_a = 20 W/(kg·K) was assumed, which is consistent with published data for fixed bed alkylation.

5.1.3.1 Effect of Inlet Temperature (Case A)

The coolant temperature was kept fixed at T_{cool} = 600 K, whereas T_0 was altered through four temperatures as follows: 610, 620, 628 (design), and 638 K. Under these circumstances, the amount of catalyst was changed in such a way that X_A = 0.99 remained steady for all the cases considered.

Table 3: Case A: Effect of T_0 (T_{eool} = 600 K, X_a = 0.99)

T_0 (K)	T_{eool} (K)	T_{hot} (K)	Sc	X_a	$W^{Re\varphi}$ (t)
610	600	646.9	0.9725	0.9900	124.6
620	600	654.1	0.9724	0.9900	128.3
628	600	672.5	0.9719	0.9900	130.2
638	600	838.7	0.9511	0.9900	131.3

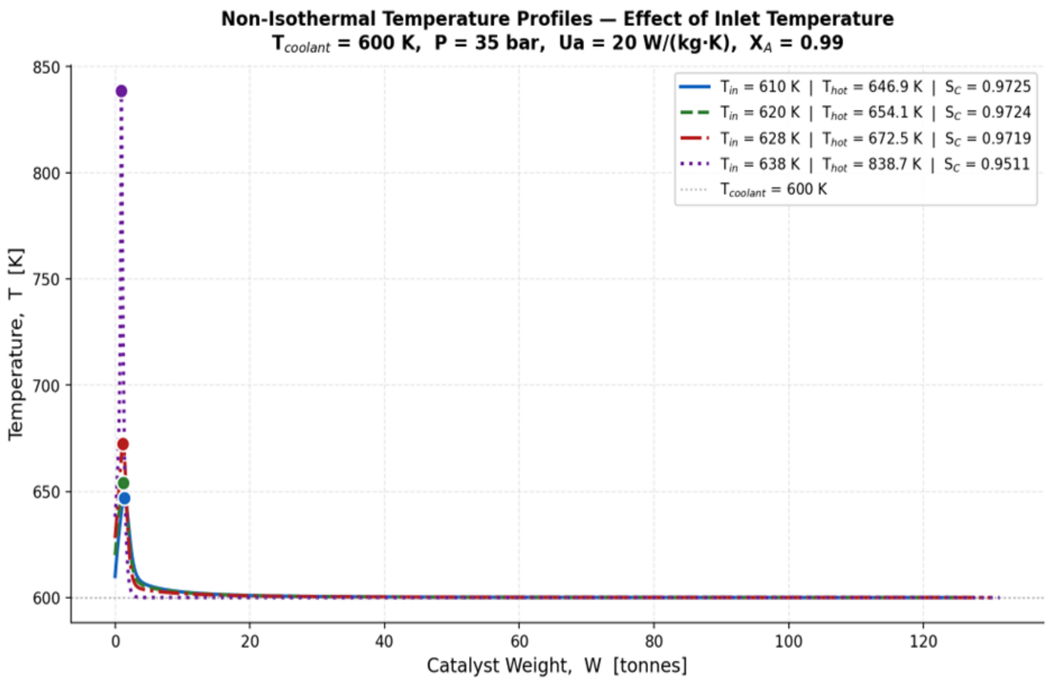

Figure 3: Temperature Profiles Along the Catalyst Bed for Four Inlet Temperatures.

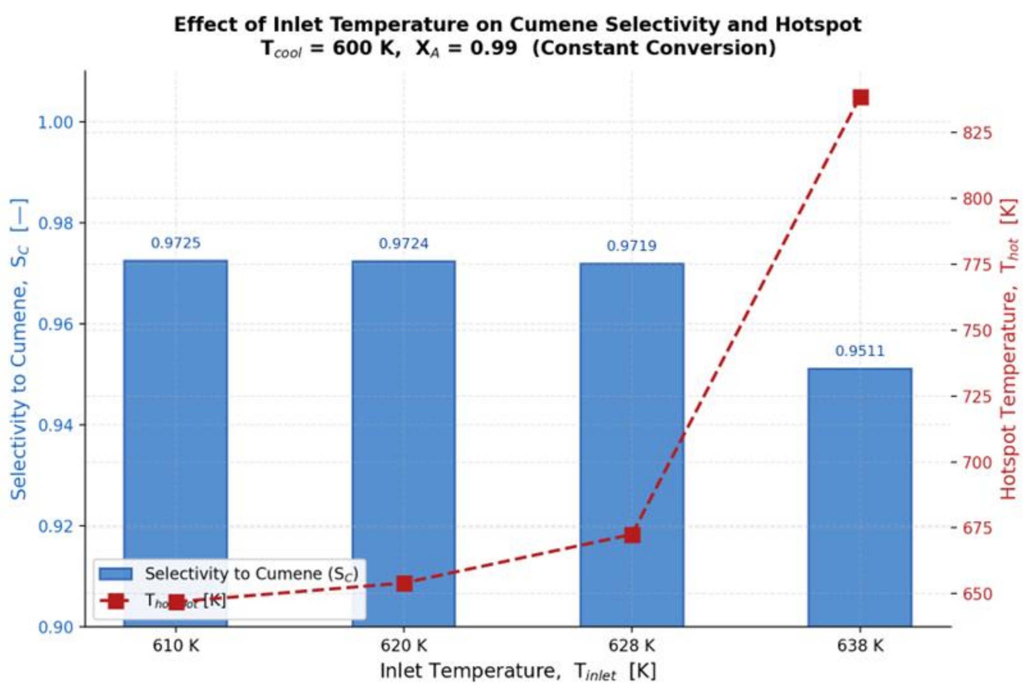

Figure 4: Cumene Selectivity and Hotspot Temperature vs. Inlet Temperature.

Hot spot temperature increases from 646.9 K to 838.7 K and Cumene selectivity reduces from 0.9725 to 0.9511 as T_0 increases from 610 K to 638 K. Such characteristics can be attributed to the high activation energy associated with the side reaction as the DIPB production is more temperature-sensitive compared to the formation of Cumene. The concentrations of reactants and heat generated are maximum near the inlet of the reactor (W_{hot} = 1-2 tons). As T_0 = 838.7 K is quite near to the maximum permissible temperature (about 800-850 K) for zeolitic alkylation catalysts, the operation conditions at T_0 = 638 K should not be used.

5.1.3.2 Effect of Coolant Temperature (Case B)

The coolant temperature, T_{cool}, was varied between 580 K, 590 K, and 600 K for a fixed value of T_0 = 628 K. Due to their tendency to generate almost run-away hotspot effects (T_{hot} > 900 K), coolant temperatures greater than 602 K were excluded at the given U_a.

Table 4: Case B: Effect of T_{cool} (T_0 = 628 K, X_a = 0.99)

T_0 (K)	T_{Cool} (K)	T_{hot} (K)	S_C	X_a	$W^{Re\varphi}$ (t)
628	580	628.0	0.9757	0.9900	272.3
628	590	628.9	0.9742	0.9900	189.1
628	600	672.5	0.9719	0.9900	130.2

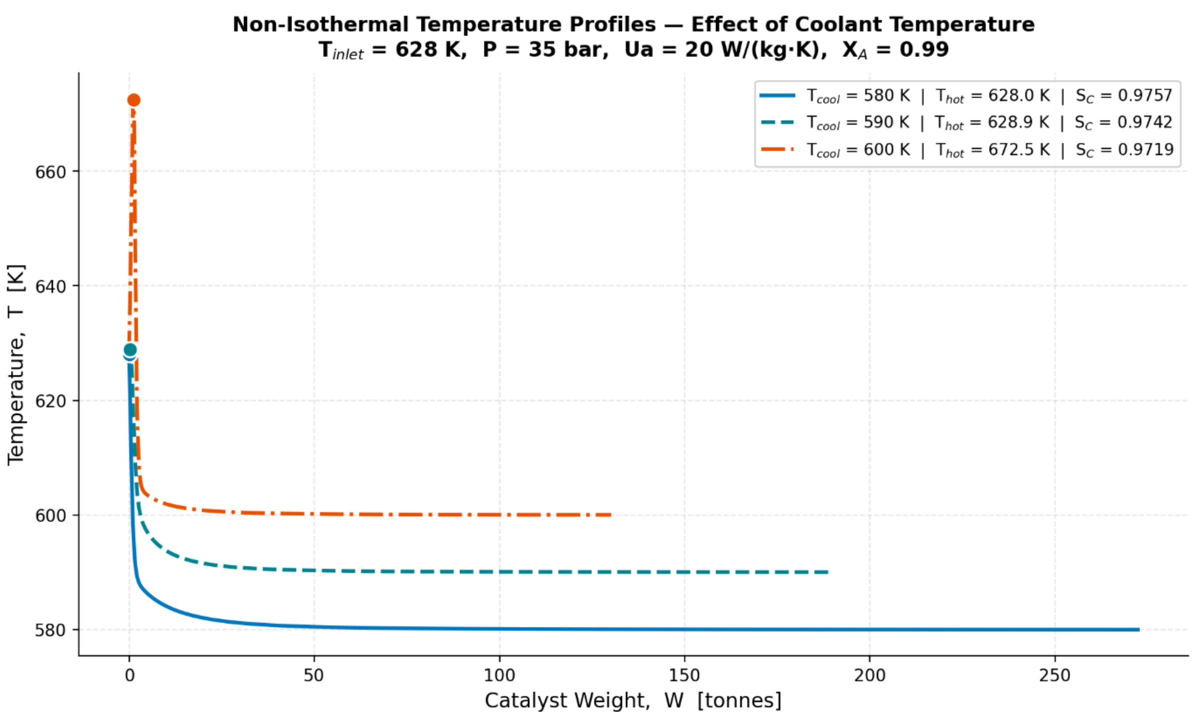

Figure 3: Temperature Profiles Along the Catalyst Bed for Three Coolant Temperatures.

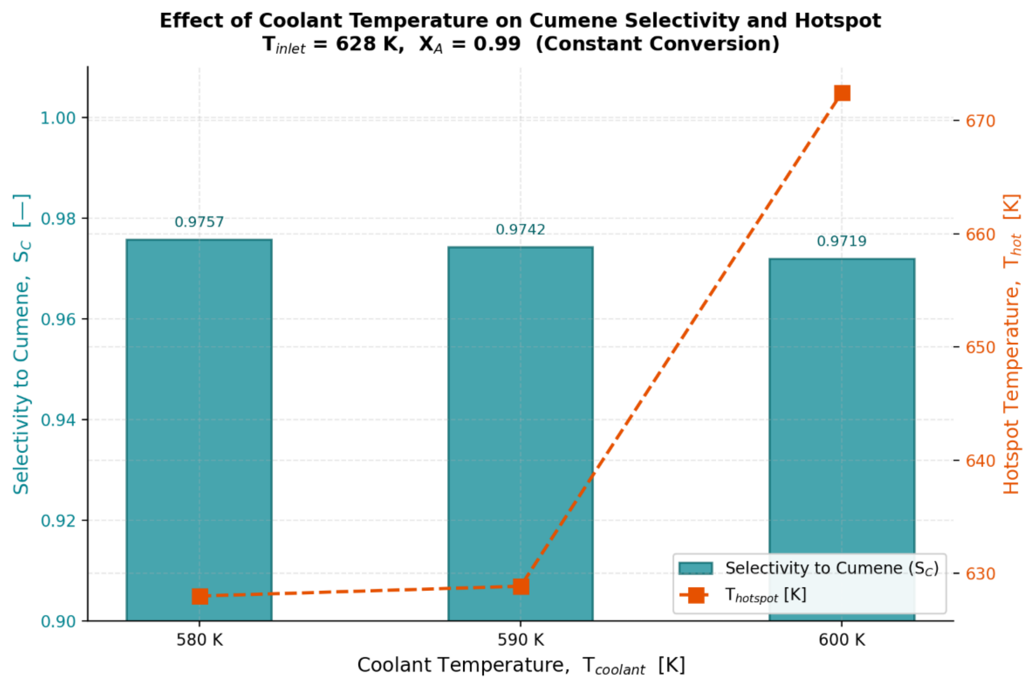

Figure 4: Cumene Selectivity and Hotspot Temperature vs. Coolant Temperature.

Selectivity rises from 0.9719 to 0.9757, while the hotspot temperature is reduced from 672.5 K to 628.0 K as a result of reducing T_{cool} from 600 K to 580 K. Nevertheless, since lower average bed temperatures mean slower kinetics, the amount of catalyst needed to attain the same conversion rate would be much higher, rising from 130.2 to 272.3 tones. The effect of the hotspot on the non-isothermal case amplifies the similar problem of selectivity-reactor size trade-off as discussed in Memo 4 for the isothermal case.

Discussion

A similar trade-off can be observed for both operational parameters: high cumene selectivity conditions (low T_0, low T_{cool}) require a larger reactor and more catalysts, while minimizing reactor size increases the hotspot temperature and reduces selectivity. Comparing the two conditions, T_0 has a stronger effect: when T_0 varies from 610 K to 638 K, S_C decreases by 2.2 percentage points, while when T_{cool} changes from 580 K to 600 K, S_C decreases only by 0.4 percentage points. This is due to the fact that the inlet conditions where the hotspot occurs depend directly on T_0. Based on the results obtained, we can conclude that the optimal operational area is T_0 = 620–628 K and T_{cool} = 590–600 K, ensuring that W_{req} remains between 128 and 130 tons, $S_C \geq 0.972$, and $T_{hot} <$ 675 K.

5.1.4 Reactor Gain & Stability Analysis

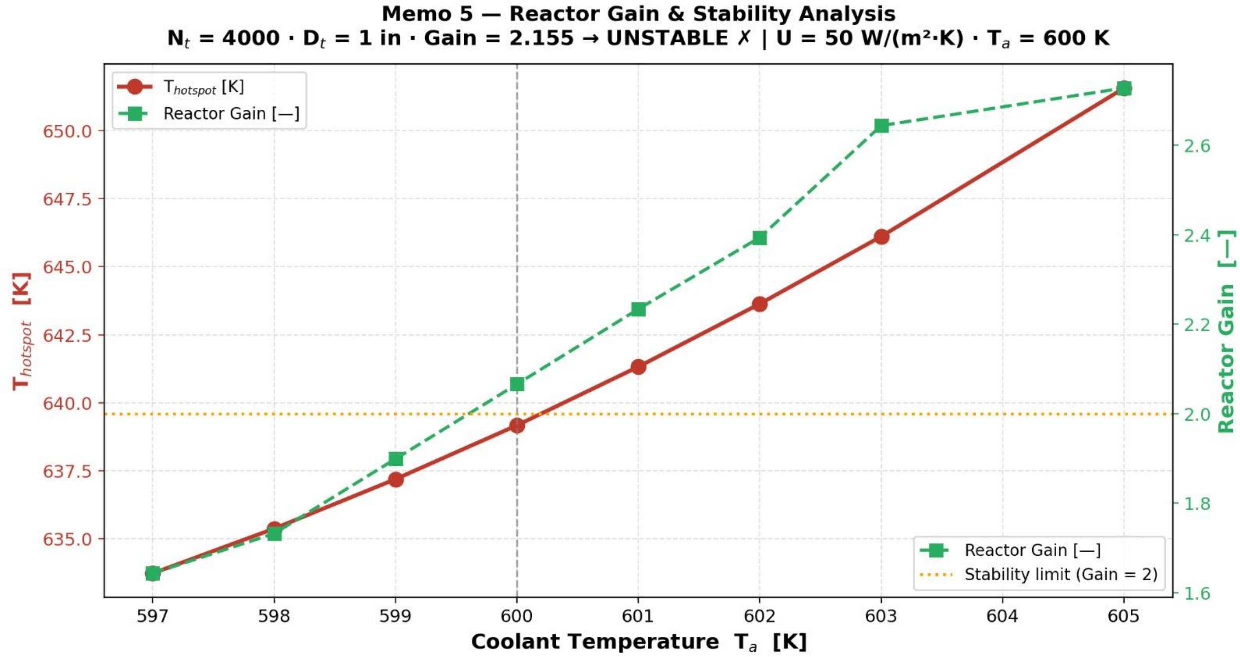

Figure 5: Reactor Gain & Stability

The objective of the reactor gain study was to ascertain the stability of the reactor under the defined operating conditions, as well as the effect that the coolant temperature will have on the hotspot temperature. The gain of the reactor is an indicator of its stability. It defines the amount by which the hotspot temperature changes with respect to a unit change in the coolant temperature.

From Figure, it can be seen that there is an increase in hotspot temperature and also in the reactor gain with increasing coolant temperature (Ta). Between approximately 634 K at Ta= 597 K and approximately 651 K at Ta= 605 K, the hotspot temperature slowly increases almost linearly. Due to reduced heat losses at higher coolant temperature, this indicates that the coolant temperature affects the reactor temperature greatly.

However, it is even more important to note that with the increase in coolant temperature, the gain of the reactor (dashed green line) sharply increases. At lower coolant temperatures (approximately 597-599K), the gain is below 2 indicating stable behavior. However, with increasing coolant temperature towards 600 K, the gain is approximately: **Gain ≈ 2.155**

At this point, the reactor will be considered dynamically unstable because this value is above the stability threshold of 2. At this point, the benefit continues to increase, and at high coolant temperatures, it exceeds 2.5. This scenario shows how one characteristic of the danger of thermal runaway is that small changes in coolant temperature will lead to considerable increases in hotspot temperature.

There are certain factors by which equilibrium between heat production and heat dissipation may explain the dynamic instability. First of all, the temperature difference for heat dissipation decreases as coolant temperatures continue to increase. Secondly, the increase in temperature also results in an increase in the reaction rate, thus increasing heat production. At this point, the reactor is unstable, and the reactor gain surpasses the critical value.

This change in behavior of operation from stable to unstable can be seen in the graph. The critical operating condition occurs when $T_a \approx 600$ K at the operating point of the junction where the system intersects the stability boundary, Gain = 2. The reactor operates in a stable manner below this critical point but becomes increasingly unstable beyond it.

This implies that the current conditions ($U = 50$ W/m²·K and $T_a \approx 600$ K) may not be suitable from an operational perspective. If the gain of the reactor exceeds two, it cannot be utilized. The temperature of the coolant needs to be reduced below the critical temperature or the heat dissipation capability increased in order for steady operation to be achieved.

Reducing the coolant temperature for an adequate temperature difference or increasing the heat transfer coefficient (U) for improving the heat loss process could be two possibilities. In this way, the gain of the reactor will be decreased, which would bring the reactor back into the stable regime of operation.

In general, from the diagram, it is clear that the sensitivity of the reactor with respect to the coolant temperature is high under the specified condition.

5.2: Interactive Slider Visualization - Discussion

An interactive tool was built to play with the non-isothermal Cumene reactor in real time. It solves the full six-equation ODE system from Part 1 — five mole balances and the energy balance — using RK4, and the plots refresh as soon as you move a slider.

Three sliders are exposed: inlet temperature T_{in} (580–680 K), HITEC coolant temperature T_a (500–650 K), and overall heat transfer coefficient U (0–300 W/m²·K). For any combination, the tool plots $T_{(W)}$ with the hotspot marked, the cumulative selectivity and conversion along the bed, and four KPIs: hotspot temperature, exit conversion X_A, Cumene selectivity S_C, and reactor gain. The stability indicator goes green for Gain < 2 and red for Gain ≥ 2.

A few things came out of the sensitivity sweep:

- **U Sensitivity**
  At baseline (T_{in} = 628 K, T_a = 600 K), U = 50 W/(m²·K) puts the reactor right on the edge with Gain = 2.18. Increasing U to 65 W/(m²·K), which is realistic with a better shell-side baffle layout, drops Gain to roughly 1.1 and pulls things into the stable region without costing much conversion. Past U = 70 W/(m²·K) the hotspot disappears, but conversion slides under 95.7% and you're spending more capital for less product. So 65 W/(m²·K) is recommended.
- **T_a Sensitivity**
  Below 580 K, selectivity is great ($S_C > 0.98$) and Gain is essentially zero, but conversion falls under 85% which is nowhere near the 99% target. Above 610 K, Gain crosses 2 and which is near the boundary. The usable window is 580–605 K: conversion around 96%, selectivity above 0.94, Gain under 2.
- **T_{in} Sensitivity**
  Increasing the inlet temperature mostly just makes the hotspot taller. Exit conversion barely improves because the byproduct reaction has a higher activation energy than the main one (Ea_2 = 120.2 vs Ea_1 = 104.2 kJ/mol), so it speeds up more with temperature. That lines up with what Memo 4 states about cooler operation favoring Cumene selectivity, and the trade-off is pretty obvious once you start dragging the slider.

One thing the selectivity panel really drives home: S_{Cumene} drops fastest right inside the hotspot, near the bed entrance. Whatever selectivity you lose there isn't coming back downstream, so the right place to fight is the hotspot itself, with cooling.

References

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